

SECTION 10 – TOOLS AND TEST EQUIPMENT

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10-1 INTRODUCTION

The following tables lists the tools and test equipment needed to install a Signa 1.5T TwinSpeed and calibrate the system.

10-2 MOVING METAL MEASUREMENT EQUIPMENT

TABLE 10-1
MOVING METAL MEASUREMENT EQUIPMENT

ITEM	GE PART NUMBER	DESCRIPTION
1	—	Magnetometer sensitive to 50 μ gauss Flux gate or equivalent Example: Barrington Instruments Ltd. Model Mag-03MC 500 μ T
2	—	Spectrum analyzer capable of moving metal measurements include the GenRad 2515, B&K 2032 or HP 3560A Dynamic Signal Analyzer.
3	—	Accelerometer capable of measuring from 0.2 Hz beyond 250 Hz.

10-3 RIGGER/CUSTOMER SUPPLIED EQUIPMENT**Note**

Table 10-2 contains an overview of equipment for initial planning purposes, refer to *Direction 2226318, 1.5T & 1.0T LCC Magnet Signa MR/i Delivery and Installation Manual* for details of equipment and requirements.

TABLE 10-2
RIGGER/CUSTOMER SUPPLIED EQUIPMENT

SUPPLIED BY	ITEM	DESCRIPTION
Rigger	1	Crane for removing magnet crate from delivery truck: 12 ton (10,909 kg)
	2	Motorized tow vehicle for pushing/pulling magnet when it is on dollies (e.g. an electric fork lift).
	3	Steel floor plates (8) to cover floors while transporting magnet, 36 in. x 12 in. x 0.25 in. (914 mm x 305 mm x 6.4 mm).
	4	Wood blocks, assorted sizes.
	5	Jacks and accessories for lifting magnet: 20,000 lbs (9091 kg) total capacity
	6	Spreader Beam
	7	For LCC magnet: Lifting straps which meet specifications in <i>Direction 2226318, 1.5T & 1.0T LCC Magnet Signa MR/i Delivery/Installation Manual</i>
	8	For TRM Installation: The following equipment to meet specifications in <i>Direction 2304881 Signa TwinSpeed Upgrade Pre-Install</i> - Forklift or crane/hoist rated for 4,000 lbs (1814 kg) - spreader beam - lifting straps
Customer	9	Equipment for off loading electronics and other miscellaneous components. (e.g. fork lift, hand trucks, straps, etc.)
	10	Panel lifters for computer flooring: Smooth floor Standard carpeted floor Level loop carpet floor Indicon Industries PL2DC; Indicon Industries PL30P; Indicon Industries SCLV1.

10-4 CRYOGENIC EQUIPMENT

TABLE 10-3
CRYOGENIC EQUIPMENT

ITEM	GE PART NUMBER	DESCRIPTION
1	46-306734G1	Low Pressure Regulator Kit (Non-magnetic gas cylinder regulator / hose assembly consisting of regulator, hose, and case) (See Note 2)
2	—	Liquid helium in non-magnetic dewars is needed for refilling the magnet (See Note 2). Refer to appropriate magnet manual for helium volume.
3	46-306717G1	Non-magnetic gaseous helium cylinder cart (See Note 2)
4	46-294705G1	Universal Fill Line Kit (See Note 2)
5	46-294511P1	250 Liter Dewar Stinger (See Note 2)
6	46-294511P2	500 Liter Dewar Stinger (See Note 2)
7	46-294512P1 (12 ft) 46-294512P2 (8 ft)	Transfer Line (See Note 2)
8	46-282336P2	Dewar stinger assembly (High Efficiency) (See Note 2)
9	46-306812G1	Dewar tube "Thumper" tool (See Note 2)
10	2253802	Burst Disk and Gasket Assembly for LCC Magnet Helium Vessel (See Note 1)
11	46-271136G1	Dewar Adapting Kit including O-ring Kit for Dewar Adapting kit (46-271135P9) (See Note 2)
12	46-271137G1	Cryogen Safety Shield Kit (Includes gloves, face shield, and safety glasses.) (See Note 2)
13	46-294804G1	Non-magnetic Aeroquip wrench Set (See Note 2) includes the following: • 46-294800G1 case for bronze wrench kit • 46-294805P1 bronze 1 5/8 in. open end wrench • 46-294805P2 bronze 1 3/8 in. open end wrench • 46-294805P3 bronze 1 3/16 in. open end wrench • 46-294805P4 bronze 1 1/8 in. open end wrench • 46-294805P5 bronze 1 in. open end wrench
14	46-281088G3	Shield Cooler Installation / Maintenance Kit (See Note 2)
15	46-265273G1	Liquid Helium Level Meter Kit (See Note 2)
16	46-306781G1	Helium Mechanical Gas Flowmeter (See Note 2)
17	46-301477G1	Cryogenic Thermometer, Lakeshore 208 Cryogenic Thermometer Kit or alternate Diode Temperature current source 46-317543G1 (See Note 2)
18	46-252210P1	3 Inch Valve Operator (vacuum break tool) (See Note 1)
19	46-294872G2	SAV-CON / Instrumentation Lead Service Kit (See Note 2)
20	46-318784G2	Shield Cooler Test Kit (See Note 2)
21	46-318696G1	Water Tee for Shield Cooler Water Samples access (See Note 2)
22	46-294052G1	Water Flow Meter Kit (for checking flow of water to Shield Cooler Compressor) (See Note 2)
23	2171219	RUO Temperature Monitor for LCC 1.5T & 1.0T Magnet (See Note 2)
Note 1 Supplied as part of Signa. 2 Supplied by GE until turnover of system to customer, then available as part of a GE Cryogen and/or Service Contract.		

10-5 INSTALLATION EQUIPMENT

TABLE 10-4
INSTALLATION EQUIPMENT

ITEM	GE PART NUMBER	DESCRIPTION
1	—	Ramp for removing cabinets from pallets for International shipments (See Note 1)
2	—	Wrecking bar
3	—	Claw hammer, 3/4 lb
4	46–271138G1	Restricted Access Control Kit. Contains two plastic warning signs for posting at site during installation and service activity.
5	46–294060G4	Mapping Fixture for field plotting equipment for Cx/LCC magnet Typically not needed to map Cx/LCC Magnet depending on proximity of steel at site, refer to Direction 2294003 Signa TwinSpeed Pre–Installation Section 4–14–1 Floors
6	2163971	Field Mapping Fixture Upgrade Kit to make kit compatible with Cx/LCC Magnets Used to upgrade 46–294060G3 to 46–294060G4
7	—	Magnet Log Book
8	—	Installation log book
9	—	4 foot or equivalent carpenter level
10	2230707	For LCC Magnet: Aluminum platform ladder, 47.5 inches (1206.5mm) (See Note 1)
11	2134776	Gradient Cable Crimper/Stripper Kit (Note 2) consisting of: • 2134586 Cable stripping tool • 2134586–2 Stripping tool replacement blade • 2134587 Cable slicer • 46–282853P1 Ratcheting crimper • 2135839 1/2 inch terminals • 2135839–2 3/8 inch terminals
12	46–301450G1	Fiber optic connector repair kit (See Note 2)
13	46–260776G3	Magnet Service Tool 750 Amp Power Supply Cabinet dimensions W x D x H is 24.5 in. x 31.5 in. x 35.75 in. (622 mm x 801 mm x 908 mm) and weight of 375 lbs (170 kg).
14	46–260776G4	Magnet Service Tool 1000 Amp Power Supply Cabinet dimensions W x D x H is 24.5 in. x 31.5 in. x 35.75 in. (622 mm x 801 mm x 908 mm) and weight of 375 lbs (170 kg).
15	46–260777G3	Shim Service Tool Power Supply Cabinet dimensions W x D x H is 24.5 in. x 31.5 in. x 35.75 in. (622 mm x 801 mm x 908 mm) and weight of 290 lbs (132 kg).
16	46–306763G1	Subminiature–D Connector Removal/Re–termination Repair Kit for Robinson–Nugent Sub–D connectors
17	46–251865G2	Field Plotting Kit – Metro Lab Teslameter
Note	1 Supplied as part of Signa.	
	2 Supplied by GE until turnover of system to customer, then available as part of a GE Cryogen and/or Service Contract.	
(Continued)		

10-5 INSTALLATION EQUIPMENT (Continued)

TABLE 10-4 (Continued)
INSTALLATION EQUIPMENT

ITEM	GE PART NUMBER	DESCRIPTION
18	46-260703G3	Magnet Ramping Equipment Kit
19	46-318833G1	Ramp Cable Holder
20	46-294998G1	Ramping Supply and Equipment kit includes the following: • 46-260703G3 Magnet Ramping Equipment Kit (see Item 18) • 46-260776G3 Magnet Service Tool Power Supply Cabinet (see Item 13)
21	46-320273G3 or G4	Non-Magnetic Tool Kit – Universal (See Note 2) Both metric and inch Non-Magnetic Tool Kits needed. May substitute both of the following kits: • 46-320273G1 Non-Magnetic Tool Kit – Metric • 46-320273G2 Non-Magnetic Tool Kit – Inch
22	2164744-4	BRM Insertion Tool includes Coil Lift Kit 2326753
23	2144093	Standard Long Room coil cart with cradle for use in 20.44 ft (6.23 m) or longer Magnet Room
24	2144093-2	Short Room coil cart with cradle for use in <20.44 to 19.6 ft (<6.23 to 5.97 m) long Magnet Room
25	2299496	Ring Alignment Tool
Note 1 Supplied as part of Signa. 2 Supplied by GE until turnover of system to customer, then available as part of a GE Cryogen and/or Service Contract.		

10-6 RF SURVEY APPARATUS

TABLE 10-5
RF SURVEY APPARATUS

ITEM	GE PART NUMBER	DESCRIPTION
26	46-265169P1	Tuned dipole antenna (two required)
27	46-265169P2	Tripod for dipole antenna (two required)
28	—	0.305 m diameter loop antenna
29	46-255835P2	Spectrum analyzer
30	46-265171P1	RF Preamplifier
31	46-265170P1	RF Linear Power Amplifier

10-7 TEST EQUIPMENT

TABLE 10-6
TEST EQUIPMENT

ITEM	GE PART NUMBER	DESCRIPTION
1	2284754	Textronic TDS3012 digital scope
2	2284763	Fluke FLK196M scopemeter for medical applications
3	46-194427P226	Dual trace oscilloscope, 100 MHz bandwidth, 2 channel, digital storage, Textronic 2232
4	46-194427P222	Dual trace oscilloscope, 350 MHz bandwidth, 4 channel, Textronic 2465
5	46-194427P284	Battery operated digital multi-meter, Fluke 87, 4.5 digits with frequency counter and capacitance
6	46-208572P9	Clamp-on Ammeter, 1-200A, AC/DC
7	46-317724P14	Attenuator/ load, 200 Watt, 30 dB attenuator
8	46-317724G2	RF Power Measurement Kit with Attenuator/ load, 200 Watt, 30 dB attenuator 46-317724P14
9	2218826	RPM 1650 Power Analyzer Kit
10	46-328143G3	Dranetz 626 analyzer kit with 1 and 3 phase modules
11	Catalog E6320DA or 46-194427P144	Densitometer
12	46-306801G1	Bell Gauss Meter Kit
13	46-194427P248	Microguard or ECOS leakage tester
14	46-306797G1	Fogg (ECG) Simulator and Memory Module
15	46-320433P1	Infrared Scanner 0 – 100° C
16	46-317830G1	Fiber Optic Light Meter Kit
17	46-294047G1	Shield Cooler Vacuum Pump Kit
18	46-251867G1	Main Vacuum Pump Down Kit
19	46-252805P2 (10 ft) 46-252805P3 (15 ft)	Nitrogen Transfill line (only needed if magnet is warmer than nitrogen temperture)

10-8 CALIBRATION TOOLS/FIXTURES

TABLE 10-7
CALIBRATION TOOLS/FIXTURES

ITEM	GE PART NUMBER	DESCRIPTION
1	46-307500G1	Longitudinal Drive Force Gauge
2	2133388-6	System Performance Test (SPT) Phantom Set for Systems with Cx/LCC (Active Shield) systems (See Note 1) consisting of the following: • 2125247-10 Nesting Plate • 2125245 Large Volume Shim Phantom Assembly • 2125244 Short Loader Assembly • 2131027-2 Daily Quality Assurance (DQA III) Phantom • 2134213 SPT Shipping/Storage Cart (Not used for Mobile Systems) • 2170481 EPI Foam Positioner • 2141454 Quick Shim Positioner
3	46-287900G3	Head TLT Sphere (46-265826G6), one 100 mm Nickel Chloride sphere (46-317586G1), and Loader (46-287899G1) See Note 1
4	46-255816G1	RF Test Cables Kit
5	46-265434G1	Magnet Rundown Unit Test Box
6	46-307164G4	1.5T / 1.0T GRAFIDY II Kit (See Note 4)
7	2234621	ECMT Kit (See Note 4)
8	46-328021G1	Enmet Oxygen Monitor Calibration consisting of the following: • 1800 PSI cylinder of 20.9% Oxygen in Nitrogen • 1800 PSI cylinder of 17% Oxygen in Nitrogen • Regulators with calibration adapter For use with portable or permanent Oxygen Monitor.
9	2106236	Portable Oxygen Monitor (Connecticut Analytical)
10	2107184	Permanent Oxygen Monitor (Enmet)
11	46-301549G1	TPS RF Test Cable/Adapter Kit
12	46-301927G1	TPS RF Service Interface Kit (See Note 2)
13	46-306712G1	Torque driver kit
14	46-260201P1	N2 Precool Syphon (only needed if magnet is warmer than nitrogen temperture)
15	46-306864G1	Magnet Helium Resistance Box Kit
16	2101360	Power Supply Calibration Kit
17	46-265826G3	Head Signal-to-noise (SNR) sphere (See Note 3)
18	46-265635G4	Body Signal-to-noise (SNR) sphere (See Note 3)
19	46-265635G2	Body T2 sphere (See Note 3)
20	46-328480G1	100 mm Nickel Chloride sphere (46-317586G1) and Universal Phantom Holder (46-328383P1) See Note 2
Note	1	Supplied as part of Signa.
	2	Supplied by GE until turnover of system to customer, then available as part of a GE Service Contract.
	3	Customer may purchase these items.
	4	Either the ECMT Kit or the 1.5T/1.0T GRAFIDY II Kit can be used to calibrate the system.

10-9 TOOL KIT

TABLE 10-8
TOOL KIT

ITEM	GE PART NUMBER	DESCRIPTION
1	—	Extension cords, with ground conductor
2	—	Power strip, grounded type, with minimum of five outlets
3	—	Soldering iron, pencil type with solder
4	—	Solder sucker
5	—	Assortment of Brady Quick labels
6	—	Micro clip leads
7	—	14 pin and 16 pin DIP clips
8	46-258218P3	Vinyl electrical tape
9	46-258218P4	Copper tape, 3 in. wide
10	46-258218P5	Copper tape, 2 in. wide
11	46-258218P6	Copper tape, 1 in. wide
12	—	Alcohol cleaning solution
13	—	Plastic or aluminum flashlight
14	—	Plastic or aluminum pen light
15	AMP No. 458994-1	Pin extractor, Universal Mate'n'Lock
16	AMP No. 305183-R	Pin extractor, M-series
17	46-237072P1	Pin extractor, Sub-D
18	46-307307G1	Crimping tool for coax cable and BNC connectors Inserts for RG8, 9, 11, 214 46-255841P103 RG58, 223 46-255841P100 RG59 46-255841P101 RG174 46-255841P102 Die Removal Tool 46-255841P201
19	—	Assorted crimp tools.
20	—	Non-magnetic level
21	—	Non-magnetic tape rule, 12 ft
22	—	Assorted drill bits
23	—	Inspection mirror
24	—	Tap set, standard, and tap handle, T-type
25	—	6 in. rule, standard and metric markings
26	—	Alignment tool (tweaker)
27	—	Hex/alignment tool
28	—	Hemostat, 5 in., curved
29	46-198094P1	Wrist grounding strap
(Continued)		

10-9 TOOL KIT (Continued)

TABLE 10-7 (Continued)
TOOL KIT

ITEM	GE PART NUMBER	DESCRIPTION
30	Xcelite 110CG	Diagonal Cutting Pliers, 4-1/2 in.
31	—	Screw Starter, aluminum or plastic shaft
32	—	Hobby and utility knives
33	—	Spring scales, 0-10 lbs and 0-50 lbs
34	46-313413P1	Extractor for 20 – 100 pin PLCC Chips